Plausible Mechanisms for Hallucination in Generative AI Systems

A Response-Production Framework

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Abstract

Hallucination in generative AI systems is commonly identified when a representational claim is contradicted by, or lacks support required from, an applicable source of truth. That determination identifies an error in the delivered response but does not explain how the system produced it. This paper develops a response-production framework for examining plausible sources of hallucination. The framework traces response production from model formation to autoregressive generation and decoding. It also examines retrieval, context construction, tools, and other deployment processes. These deployment processes may operate before, during, or after generation. The framework organizes plausible error-generating pathways into three interacting mechanism families: Model Approximation, Autoregressive Realization, and Deployment Mediation. By relating these families within one production structure, the framework supports evaluating and managing hallucination at the level of a configured system performing a specified task rather than by model name alone. Existing evidence supports particular pathways and interactions to different degrees; comparative and factorial tests are needed to evaluate the integrated framework.

Keywords: hallucination; generative AI; large language models; response-production framework; model approximation; autoregressive realization; deployment mediation.

A. Introduction

Hallucination is widely recognized as a limitation of generative AI. Research has developed definitions, benchmarks, detection methods, and mitigation strategies across natural-language-generation tasks ([1], [2]). Most of this work asks whether a response contains an incorrect or unsupported claim and how often such claims occur. We take that determination as given and ask a different question: where in response production might an identified hallucination have arisen?

Answering this question requires examining the process that produces the delivered response. A trained model reflects selected data, objectives, architecture, capacity, optimization, and post-training. At deployment, the system may retrieve information, construct a context, call tools, and incorporate system instructions and prior interaction history. The model then generates conditional predictions that a decoding procedure realizes as one response path, which may be modified again before delivery. An error observed at the end of this sequence may therefore reflect more than the trained model alone.

Existing research provides important evidence about many parts of this process. Surveys organize hallucination by task, output type, contributing factor, detection method, and mitigation strategy ([1], [2]).

More recent reviews distinguish source faithfulness from world factuality in evaluation ([3]) and organize proposed causes across stages of the LLM development lifecycle ([4]). These classifications clarify evaluation targets and identify conditions associated with different parts of model development and use. They do not, by themselves, trace how those conditions combine within one configured system to produce a delivered response. We build on that work by relating model formation, response realization, and deployment to the production of one delivered response. The contribution is not simply a new set of stage labels. The framework examines interactions among processes operating in different parts of response production. These processes may correct, transmit, or amplify one another's effects. Because the delivered response reflects those interactions, hallucination should be evaluated at the level of a configured system performing a specified task rather than by model name alone.

We organize plausible sources of hallucination into three mechanism families. *Model Approximation* concerns error-generating pathways associated with forming a finite predictive model from bounded information through particular design and training choices. *Autoregressive Realization* concerns the sequential process through which conditional predictions and decoding produce one response path. *Deployment Mediation* concerns retrieval, context construction, tools, interaction history, and post-generation processes that shape what the model receives and what the user ultimately sees.

The three families are organized around principal parts of response production, but they are neither strictly chronological nor independent compartments. Model Approximation centers on conditions inherited from model formation. Autoregressive Realization follows the sequential production of a response path. Deployment Mediation can operate before realization through retrieval and context construction, during realization through tools or repeated cycles, and after realization through output processing. The same observed error can consequently arise through different paths, while an intervention in one part of production may change the effect of another. This interaction makes the configured system performing a specified task—not a model name alone—the appropriate unit of analysis for interpreting hallucination rates.

The paper contributes an organizing framework rather than a complete causal theory. It uses response production as the common structure for relating pathways studied separately in the literature, connects those pathways to evaluation and management of configured systems, and derives testable hypotheses for comparative research. The framework does not determine the source of an individual error from the response alone. It identifies where evidence should be sought and how conditions at different stages may be examined together.

The paper proceeds as follows. We begin in Section B with the response-production sequence. Sections C through E then examine Model Approximation, Autoregressive Realization, and Deployment Mediation, respectively. Section F brings the three families together by examining their interactions. Section G turns to evaluation, management, and future research before discussing the framework's limitations and conclusions.

B. From Model Formation to Delivered Response

This section presents the response-production sequence used to organize the analysis. In this paper, a configured system means the trained model together with the deployment resources, context-construction process, decoding settings, tools, and output controls used for a specified task. The system constructs an operative context from its available inputs, generates and decodes conditional predictions, and may apply post-generation processes before delivering a response. Figure 1 summarizes this sequence, identifies where each mechanism family principally operates, and shows why processes in different parts of production may interact.

We use the term mechanism family to group plausible error-generating pathways around a principal part of this sequence. The families provide an analytical ordering rather than three strictly chronological stages: Deployment Mediation may surround or recur through Autoregressive Realization. The families are also not mutually exclusive, because a process in one part of production may correct, transmit, or amplify an error-generating condition in another. Nor can the responsible production path be inferred from the delivered response alone.

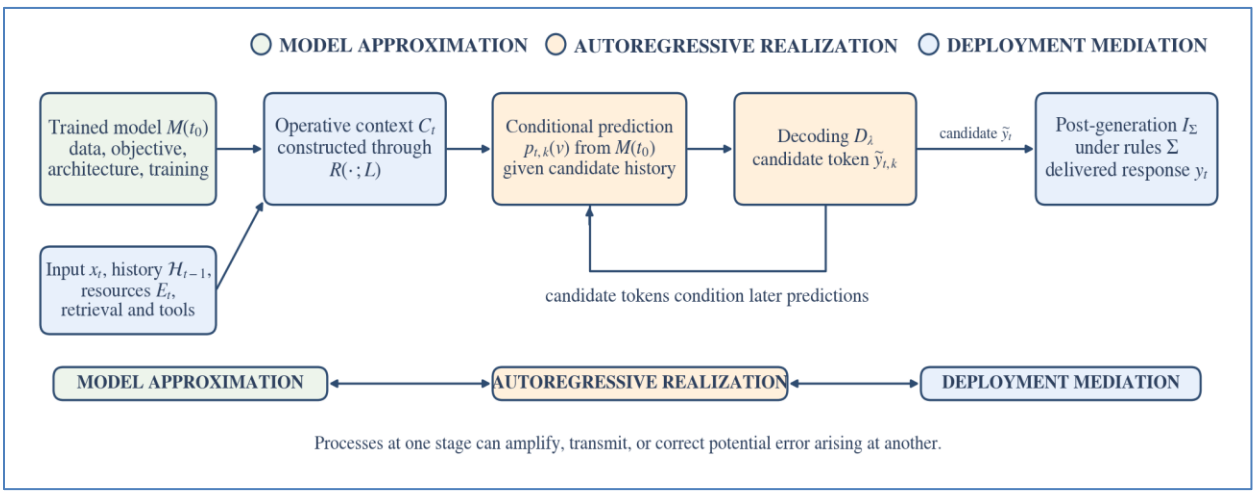

Figure 1. Response production and the three mechanism families. Each family has a principal locus, but Deployment Mediation may operate before, during, or after realization, and processes across the three families may interact.

B.1 The trained model

To describe a particular interaction, let $M(t_0)$ denote the trained model used at response time t. The time t_0 identifies the latest point at which information could have entered the model through the training and post-training processes that produced it. This is the model's temporal information boundary. The model also has a coverage boundary because it reflects only the information represented in the selected data and learning process. Together, these boundaries distinguish what the trained model could plausibly have learned from what must be supplied later through deployment.

Many contemporary language models use Transformer architectures descended from the attention-based structure introduced by Vaswani et al. ([5]). Text is divided into tokens and processed through repeated layers of attention and feed-forward transformations. Pretraining adjusts model parameters to predict subsequent tokens in large text corpora, while post-training further shapes instruction following and response behavior. The trained predictor available at response time is the framework's endpoint for model formation; earlier operations are represented through their effects on that predictor.

B.2 Operative context

Let x_t denote the current task input, including the applicable current system and user instructions, $\mathcal{H}_{t-1}$ the prior interaction history, and E_t external resources potentially accessible at time t. A context-construction process R, operating under an effective constraint L, produces the operative context:

$$C_t = R(x_t, \mathcal{H}_{t-1}, E_t; L). \qquad (1)$$

The operative context is the supplementary information actually provided to the model for generation. The construction process may retrieve, select, omit, compress, order, or label information. The current task input has two roles in this sequence: it guides the construction of supplementary context through Equation (1), and it is also supplied directly to generation in Equation (2). Equation (1) therefore distinguishes information that the deployed system could access from information actually supplied to the model.

The effective constraint L represents practical limits on context construction, such as context length, latency, permissions, or system policy.

B.3 Conditional prediction and realization

At token position k, the trained model $M(t_0)$ produces a conditional distribution over the vocabulary V:

$$p_{t,k}(v) = P_{M(t_0)}\big(\tilde{y}_{t,k} = v \mid x_t, C_t, \tilde{y}_{t,1:k-1}\big), \qquad v \in V. \qquad (2)$$

The distribution depends on the model, the current task input supplied directly to generation, the operative context, and the candidate-response path already generated. A decoding procedure D_λ, operating under settings λ, selects the next candidate-response token:

$$\tilde{y}_{t,k} \sim D_\lambda\big(p_{t,k}\big). \qquad (3)$$

In plain language, the model ranks possible next tokens, and decoding converts that ranking into one realized candidate-response token.

Repeating this operation produces a candidate response $\tilde{y}_t$. Because every selected token enters the candidate-response history in Equation (2), the response path depends on earlier selections. A deployed system may then use a post-generation implementation process I_Σ to apply rules and controls Σ:

$$y_t = I_\Sigma(\tilde{y}_t, x_t, C_t; \Sigma). \qquad (4)$$

where y_t is the response delivered to the user. Equations (1)–(4) provide a common scaffold for separating accessible information from operative context, conditional prediction from realized selection, and the candidate response from the response ultimately delivered. They organize the production analysis; they are not a causal model of hallucination.

Model Approximation principally concerns $M(t_0)$; Autoregressive Realization concerns the conditional distribution, decoding, and generated path; and Deployment Mediation principally concerns R, external resources, interaction history, tools, and I_Σ operating under Σ.

Equations (1)–(4) describe one response-production cycle. In systems that use repeated tool calls, search, test-time computation, or iterative revision, the cycle may recur: the output of one cycle becomes part of the operative context for the next. This recurrence is one reason Deployment Mediation cannot be assigned to a single chronological stage. It creates additional production paths and interactions, but the same three-family structure can still be used to examine them.

C. Model Approximation

Model Approximation concerns error-generating pathways inherited from the formation of the trained model. A generative AI model is not a truth set, a database, or a lossless copy of its training corpus. It is a finite predictive model formed from selected data through particular learning objectives and design choices. We examine three pathways through which that approximation can contribute to later error: bounded information, a predictive objective that is not identical to factual verification, and limits in how factual relationships are represented and recovered.

C.1 Bounded information

The first pathway is bounded information. Its clearest form is temporal. A model completed at t_0 cannot acquire through that completed training process facts that arise later. If an officeholder changes, a company reports new earnings, or a stock price moves after t_0, the later fact cannot be part of the original trained model. It can reach the system only through a later model update or a deployment process such as retrieval or tool use. Studies of current knowledge find that models struggle with changing information and that their effective knowledge can predate a nominal training cutoff ([6], [7]).

Bounded information also has a coverage form. A fact may have existed before t_0 but still fall outside what the model learned reliably. It may not have appeared in the selected training data, or the available examples may not have expressed the relationship clearly and consistently enough for the model to recover it later. This coverage boundary matters for private, specialized, or task-specific information as well as for facts that are rare or inconsistently represented in public text. Evidence across multiple question-answering datasets, pretraining corpora, and model sizes shows that factual accuracy can be strongly related to the number of relevant pretraining documents, while retrieval augmentation reduces that dependence ([8]). This finding supports a bounded coverage pathway while also showing how Deployment Mediation may compensate for it.

Neither form of boundary determines the delivered answer. When required information is absent from, or weakly represented in, the trained model, the system may still qualify its response, state that more information is needed, retrieve an applicable source, or abstain. The pathway instead identifies a basic limitation: some tasks require information that the trained model alone cannot be expected to provide. Whether that limitation becomes a hallucination depends on what happens during deployment and response realization.

C.2 Predictive objective and factual adequacy

The second pathway reflects our analytical interpretation of the relationship between the learning objective and factual accuracy. The standard language-model objective rewards prediction of tokens found in training text; it is not itself a procedure for verifying every proposition implied by a possible response against an authoritative source of truth. Textual probability and factual truth overlap because true relationships appear in language, but the two are not identical. A widely repeated misconception may be easy to predict, while a true but rarely stated fact may be difficult to recover. TruthfulQA provides evidence that models can reproduce common human falsehoods and that greater scale alone does not necessarily improve truthfulness on questions designed to elicit those misconceptions ([9]).

Post-training can change these tendencies by shaping how the pretrained model follows instructions and selects among possible responses. It can improve behavior, but it does not turn next-token prediction into factual verification. Controlled evidence also shows that fine-tuning on factual knowledge not already represented by a model can increase hallucination under some conditions ([10]). The relevant point is not that post-training generally worsens factuality. It is that the behavior of the resulting model reflects both pretraining and later training, and neither process guarantees that the continuation favored in a particular context will be factually correct.

C.3 Representation and recovery

The third pathway concerns whether factual relationships are represented and recovered reliably within a finite model. Exposure to a fact during training does not ensure that the trained model will reproduce it when a later task requires it. Controlled memorization studies find that factual recall varies with model scale, training duration, data quality, fact frequency, compatibility among facts, and training order ([11], [12]). A separate study introducing the term *knowledge overshadowing* suggests that a dominant association can suppress a less prominent one during generation ([13]). These findings concern particular experimental settings, but they illustrate why information present during training may still be represented or recovered unreliably.

Capacity, training conditions, competing associations, architecture, optimization, and other design choices can all shape this representation-and-recovery pathway. This pathway differs from the objective mismatch discussed in C.2. C.2 asks why predictive preference need not coincide with factual truth; C.3 asks why a factual relationship that was available during training may not be represented or recovered reliably in a particular context. The empirical success of a design does not ensure that every qualification, time reference, source identity, or factual relationship will be preserved. At the same time, an error produced by a Transformer does not show that attention, or any other single feature, caused it. The narrower claim is that the trained model is a design-contingent approximation whose construction affects which factual relationships it can recover reliably.

D. Autoregressive Realization

Autoregressive Realization concerns how conditional predictions become a completed response after the trained model and operative context have been set. The model does not produce the response all at once. At each position, it predicts possible next tokens; decoding realizes one selection; and that selection becomes part of the context for the next prediction. We examine three consequences of that process: predictive uncertainty and decoding choice, the relationship between response length and error, and propagation along the realized response path.

D.1 Predictive uncertainty and decoding

The first pathway begins with the conditional distribution in Equation (2). That distribution ranks possible next tokens; it is not a direct estimate of whether a complete factual claim is true. A concentrated distribution can favor a false continuation, while a diffuse distribution may simply reflect several valid ways to express the same proposition. Confidence in the next-token prediction and confidence in the factual correctness of the eventual claim are therefore not the same.

Even so, variation among possible generations can provide evidence about factual-error risk. Research on semantic entropy finds that differences across semantically distinct responses can help predict some hallucinations ([14]). This makes uncertainty useful as a screening signal: it can identify responses that may warrant review. It does not, by itself, show that a response is wrong or explain where an observed error arose.

Decoding is another determinant of realization because it specifies how the system moves from each conditional distribution to a selected token. Greedy decoding deterministically selects the highest-ranked option, whereas sampling procedures may reshape or restrict the distribution before making a stochastic selection. Deterministic decoding removes sampling randomness, but it cannot make an incorrectly favored continuation true. Conversely, a factuality-oriented inference-time decoding intervention has improved factuality on the models and tasks examined without changing model parameters ([15]). The realized error rate can therefore depend not only on the trained model, but also on the rule used to convert its predictions into a response.

D.2 Response length

Response length can affect hallucination in two different ways. The first is a simple increase in opportunities for error. A longer response usually contains more factual claims, so the chance that at least one claim is wrong can rise even if each additional claim is no less reliable than the earlier ones. We refer to that relationship as cumulative exposure. The phrase is descriptive rather than an established technical term.

The second relationship concerns the reliability of later content. A model may state its most strongly represented facts first and draw on weaker or more specific associations when asked to continue. One recent study calls this pattern *facts exhaustion* and finds it to be the principal explanation for declining factuality in the models and tasks examined ([16]). Because the term and result come from that particular study, they should not be treated as a general law of long-form generation.

The distinction matters because the two relationships make different claims. *Cumulative exposure* says that more claims create more chances for error; it does not require later claims to become less reliable. *Facts exhaustion* says that reliability itself may decline as generation reaches weaker factual associations. The distinction also implies different controls. Concise responses can reduce the number of opportunities for error, while requests for extensive detail may require retrieval or verification if later claims are expected to draw on weaker knowledge.

D.3 Path dependence and propagation

Each selected token enters the history $\tilde{y}_{t,1:k-1}$ used to produce the next prediction. If a response first selects the wrong person, date, place, or premise, later generation may elaborate on that selection. The resulting text can be coherent with the earlier response while remaining false relative to the applicable source of truth.

Experiments on hallucination snowballing show that models can generate false explanations after an incorrect initial answer, including claims they can reject when those claims are evaluated separately ([17]). This finding supports a propagation pathway: an early erroneous selection changes the conditioning information for what follows and can produce dependent errors. It does not imply that every long response deteriorates or that every later error originates in an earlier one.

Related evidence from neural machine translation connects propagation to a mismatch between model training and realization. Under teacher forcing, the model is trained on correct prior tokens, whereas inference conditions later predictions on the model's own selections. Experiments in neural machine translation under domain shift found that exposure bias contributed to hallucination and that a sequence-level training intervention reduced it ([18]). Although this result is specific to the translation settings examined, it illustrates how model formation, generated history, and later error may be connected.

E. Deployment Mediation

Deployment Mediation concerns the processes that connect a trained model to the task it performs in use. These processes may supplement the model with needed information or capabilities, determine what enters the operative context and how it is presented, govern how tools are executed and incorporated, and transform the candidate response before delivery. They may therefore prevent an error, create one, or change whether an error arising elsewhere reaches the user. We follow these relations from task-adequate supplementation to the delivered response.

E.1 Information supplementation

The first question is whether deployment provides information and capabilities adequate for the specified task. Retrieval-augmented generation can combine learned model representations with information obtained from external sources ([19]). Retrieval may supply facts that arose after t_0, private organizational information, designated documents, or specialized evidence that model formation could not provide. Tools may similarly supply calculations, database results, observations, or current market information at response time.

When the task depends on such information or capabilities, their absence or inadequacy creates a deployment-level pathway to error. A static model may be asked for information it could not have learned, or a system may provide a source that is outdated or unsuitable for the task. Supplying access to an appropriate resource can close this gap without changing model parameters. Access alone, however, does not ensure that the right material will be retrieved and placed in the operative context.

E.2 Retrieval and context construction

Information does not influence generation merely because it exists in an accessible resource. Retrieval may omit the relevant source, return an outdated document, or select a textually similar but inapplicable passage. Even correctly retrieved material may be truncated, compressed, reordered, or stripped of temporal and source qualifications before it enters the operative context C_t.

Retrieval and context construction are related but distinct. Retrieval identifies information for possible use, while context construction determines what the model actually receives and in what form. Evidence quantity and ordering have affected correctness in current-knowledge question answering ([7]). A retrieval capability becomes valuable for the task only when applicable material is retrieved and presented in a suitable form.

Context construction can also transmit an error already present in the task input or supplied material. Experiments with false-premise questions show that models may accept an erroneous premise and elaborate it into a hallucinated response. This behavior was associated with a small subset of attention heads, and constraining those heads during inference improved performance in the reported models and tasks ([20]). The false premise enters through Deployment Mediation, but the resulting response depends on how the model processes and realizes that input; a misleading premise does not mechanically determine hallucination.

E.3 Presence and effective use

Correct information can enter the operative context C_t without governing the response. The model may rely on a stronger learned association, fail to connect the evidence to the question, or blend conflicting sources. Research on long contexts also finds that performance can vary with the position of relevant information, including lower performance when decisive evidence appears in the middle ([21]). In another study, the ratio of attention to supplied context relative to previously generated tokens provided a useful signal for contextual hallucination, and classifier-guided decoding reduced hallucination in the reported settings ([22]). We treat this result as evidence consistent with an effective-use pathway, not as evidence that an attention ratio universally causes contextual hallucination.

This distinction between presence and effective use marks an interaction rather than another retrieval step. Deployment controls which evidence is selected, labeled, ordered, and placed in the operative context. We therefore begin the discussion in Deployment Mediation. Whether that evidence governs the response must then be investigated across Model Approximation and Autoregressive Realization. An error can arise because the context is inadequate, because adequate evidence is not used effectively, or because both conditions occur together.

E.4 Tools and post-generation processes

Providing a tool does not ensure that it will be used correctly. The system must select an appropriate tool, form a suitable query, receive a reliable result, and incorporate that result into the response. For example, a system may call the correct market-data service but request the wrong date, misread the returned field, or attach the result to the wrong security. Tool execution and incorporation can therefore create error even when the necessary capability is available.

Deployment continues after the model produces a candidate response. The system may verify, filter, shorten, translate, or rewrite the candidate before delivery. These operations can remove an error, but they can also introduce one or remove a necessary qualification. Deployment Mediation therefore includes both the processes that supply and execute task-relevant resources and the post-generation operations that determine the response ultimately seen by the user.

F. Interaction Across Mechanism Families

Interaction has four principal consequences in this framework. A process in one family can correct or compensate for a limitation in another, but it can also transmit or amplify that limitation. Conditions elsewhere in production may change the effect of an intervention, and several production paths may remain consistent with the same observed error. These consequences explain why the families should be distinguished analytically but not evaluated as independent compartments.

Correction and amplification can be seen in the same example. Suppose a user asks who currently holds an office that changed after t_0. The former officeholder may remain a strong learned association because of Model Approximation. If Deployment Mediation supplies the current official record and presents it clearly, the system may correct that limitation before delivery. If it fails to retrieve the record or places it ineffectively, Autoregressive Realization may select the former name and generate policy details consistent with that selection. The initial limitation is then transmitted and amplified into additional false claims.

Because these processes interact, the effect of an intervention is cross-dependent. Retrieval does not have one fixed effect independent of the model and decoding process: it helps only if applicable evidence reaches the operative context and influences realization. Verification likewise depends on what it checks and how its result is used. A change in one family may have little effect, or even an adverse effect, when conditions in another family remain unsuitable for the task.

Existing studies illustrate this cross-family dependence. In retrieval-augmented generation, conflict between retrieved evidence and parametric knowledge can produce outputs that ignore or blend the two sources ([23]). Context-sensitive decoding interventions have improved faithfulness without retraining the model ([24], [22]), while false-premise experiments show how erroneous deployment input can interact with model processing during realization ([20]). These studies examine particular systems and interventions rather than the three-family framework as a whole. Within the settings examined, their results support evaluating learned representation, supplied context, and response realization jointly rather than as independent elements.

Interaction also creates observational equivalence: the same delivered error can be consistent with different production paths. An incorrect date might reflect a weak learned representation, an outdated retrieved source, selection among competing continuations, or alteration during post-generation processing. The response alone does not reveal which path produced it. For diagnostic purposes, this means evaluating the configured system and its task rather than inferring the source of error from a model name or delivered response alone.

G. Discussion and Conclusion

G.1 A response-production approach

Observational equivalence has a direct operational implication. After a hallucination has been identified, the evaluator should document the configured system and specified task, then examine evidence from the parts of production that could have generated or transmitted the error. Relevant evidence may include available and retrieved sources, the operative context, decoding settings, tool traces, intermediate responses, and output controls. The procedure considers all production paths supported by the available evidence and leaves the path undetermined when that evidence is insufficient.

Management can then match controls to the pathways relevant to the task. Retrieval can supply absent or outdated information; concise response requirements can reduce cumulative exposure; and verification can detect errors arising at several points. No control works equally well in every setting. Lowertemperature decoding cannot supply information that is absent, and retrieval cannot correct an error when the retrieved material is outdated, misleading, excluded from the operative context, or used incorrectly.

The same procedure also makes interactions among controls visible. Expanding accessible evidence will not help if context construction excludes it, and a reliable tool result will not help if the system incorporates it incorrectly. A stronger model may still answer a current-information question incorrectly if the deployment supplies no current data. Evaluation should trace the evidence available at each relevant part of production, while management should test whether the combined controls are adequate for the specified task.

G.2 Interpreting temporal hallucination more appropriately

Consider a system built on a model whose training and post-training were completed in February. The system is deployed without access to current market data. On June 3, a user asks for Company X's stock price on May 31, and the system returns an incorrect price. The response may be classified as a hallucination, but that classification identifies only the error. The May 31 price fell outside the model's temporal information boundary, and the deployed system had no current-data capability through which to obtain it.

The example changes how the result should be interpreted. It would be misleading to treat the wrong price as evidence that the model failed to learn or retain a fact that did not yet exist when model formation ended. The failure instead shows that the configured system was not equipped for a task requiring current market information. Comparisons among systems answering such questions should therefore account for their access to applicable current data.

This interpretation does not excuse the wrong answer. A system designed for current-price questions should consult a credible data source, recognize when current information is unavailable, qualify its answer, or abstain rather than fabricate a price. The management implication is to keep the deployment state adequate to the information demands of the task—for example, through a reliable tool connection or a robust and well-governed retrieval process.

The same principle applies beyond finance. Law, officeholders, product specifications, organizational policies, medical guidance, and continuing-project state can all change after model formation. Periodic model updates can narrow some gaps, but static retraining cannot keep every deployed system aligned with every changing task. Information sources, indexes, tools, permissions, context policies, and output controls therefore require continuing maintenance.

G.3 Testable implications and existing evidence

We summarize the framework's testable propositions in Table 1. H1 through H5 describe relationships associated with particular pathways; H3 contains two subparts because response-level exposure and later claim-level factuality require different tests. H6 states a family of factorial interaction propositions rather than one universal effect. We derive these propositions from the framework; they are not empirical findings of this paper. Each test should specify whether its outcome concerns conformity to a supplied source, factuality relative to external knowledge, or both, because those evaluation targets can diverge

([3]). The cited studies report results for particular models, tasks, and configurations and provide evidence consistent with selected premises or relationships. Testing the broader framework will require comparative and factorial designs that vary conditions across mechanism families.

Table 1. Testable propositions mapped to the response-production framework

Testable proposition	Principal mechanism family	Testable implication and evidence relationship
H1—Temporal supplementation	Model Approximation × Deployment Mediation	For facts established after t_0, supplying applicable current evidence should reduce hallucination relative to an otherwise equivalent unaugmented condition when that evidence reaches the operative context and can influence realization. The size of this effect may depend on context construction and realization conditions. A temporal-alignment study supports the premise ([6]); search augmentation improved current-knowledge performance in another study ([7]).
H2—Realization sensitivity	Autoregressive Realization	Holding the model, input, and operative context constant, a factuality-relevant decoding change that alters realized selections should change the frequency or composition of hallucinated claims. A decoding intervention provides evidence that decoding can affect factuality in the models and tasks examined ([15]).
H3a—Cumulative exposure **H3b—Later-claim factuality**	Autoregressive Realization	H3a: When added length introduces additional factual claims, the response-level probability of at least one error should rise even if claim-level reliability is unchanged. H3b: In some tasks and models, later claims should show lower claim-level factuality. One study reports evidence for the second relationship and calls its proposed explanation *facts exhaustion* ([16]).
H4—Propagation	Autoregressive Realization	When an early erroneous commitment conditions later generation, correcting or preventing that commitment should reduce dependent errors later in the response. Snowballing results support the propagation premise motivating this intervention test ([17]). Neural-machine-translation evidence also links exposure bias to hallucination under domain shift in the settings examined ([18]).
H5—Context mediation	Deployment Mediation, with response use observed	H5a: Holding the accessible evidence set constant, changing retrieval or selection should affect what evidence enters the operative context. This mediation outcome should be measured separately from any resulting change in response factuality or source conformity. H5b: Holding the evidence included in the operative context constant, changing its position or presentation should affect response factuality or source conformity if presentation affects effective use. Existing studies support evidence-quantity, ordering, positional, and controlled evidence-perturbation tests ([7], [21], [25]); compression and labeling remain proposed extensions.
H6—Interaction family	Across the three mechanism families	Factorial tests should examine whether the effect of an intervention in one family depends on a controlled condition in another. For example, the effect of retrieval quality may vary with evidence position or with a decoding rule designed to increase contextual faithfulness. Existing results are consistent with cross-family dependence ([23], [24], [20], [22]); systematic factorial testing remains needed.

Existing RAG evaluation work suggests practical components for implementing Deployment Mediation and configured-system tests. Retrieval Augmented Generation Assessment (RAGAs) treats retrieval, contextual faithfulness, and generation as distinct dimensions of a RAG pipeline ([26]). Human-Calibrated Automated Testing (HCAT) focuses on RAG systems and combines structured test generation, system-level evaluation, human calibration, robustness testing, and weakness analysis ([27]). ReEval provides a further operational precedent by perturbing evidence in prompts to generate adversarial test cases for retrieval-augmented systems ([25]). Context-relevance measures can support retrieval and selection tests; faithfulness and answer-relevance measures can support presentation-and-use tests; and structured evidence perturbation can compare how complete configurations respond to controlled changes. These works provide operational precedents, not validation of the broader response-production framework.

G.4 Limitations

The three families are broad analytical categories, not an exhaustive taxonomy. New architectures or deployment designs may require refinement, and some processes necessarily span more than one family. Evidence placement, for example, is a deployment decision, while the model's response to that evidence also reflects model formation and autoregressive realization. The categories are useful because they organize inquiry, not because every process fits exclusively within one of them.

The relative frequency and importance of the three families are empirical quantities that are likely to vary by model, task, and configuration. Selection of the governing source of truth precedes this framework. We begin after a hallucination has been identified and organize the production processes through which the erroneous response may have arisen.

G.5 Conclusion

Hallucination is observed in a delivered response, but that response emerges from processes organized here into Model Approximation, Autoregressive Realization, and Deployment Mediation. These families correspond to principal parts of response production rather than three strictly chronological stages. Organizing plausible error-generating pathways in this way connects findings often studied separately and makes cross-family interaction part of the analysis.

The resulting framework changes both evaluation and management. Evaluation should describe the configured system and the task it was asked to perform rather than assign every hallucination to the model by default. Management should examine which parts of production can supply missing information, prevent an erroneous path, or detect an error before delivery, and should keep those capabilities aligned with changing task requirements.

Existing studies provide different degrees of support for pathways involving temporal and long-tail information, predictive uncertainty, decoding, response length, propagation, retrieval, context use, false premises, and conflict between retrieved and parametric knowledge. Future comparative and factorial tests can examine how these pathways combine, where the three-family structure requires refinement, and how the importance of each family changes across configured systems and tasks.

The central proposition is simple: to understand and manage an identified hallucination, examine the sequence that produced the response, not only the name of the model that participated in it.

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